

Question			Answer/Indicative content	Marks	Part marks and guidance											
1		i	$\left(\frac{5}{6}\right)^2 \times \frac{1}{6} = \frac{25}{216} (= 0.116)$	M1	For 5/6 (or 1 – 1/6) seen	If extra term or whole number factor present give M1M0A0										
		i		M1	For whole product											
		i		A1	cao	Allow 0.12 with working										
					Examiner's Comments											
					This was well answered with the most candidates gaining all 3 marks. Some calculated $(1/6)^3$ and so scored 0 and some just scored 1 for $\frac{1}{6}$.											
		ii	$1 - \left(\frac{5}{6}\right)^{10} = 1 - 0.1615 = 0.8385$	M1	For $(5/6)^{10}$ (without extra terms)	Allow 0.838 or 0.839 without working and 0.84 with working. For addition $P(X = 1) + \dots + P(X = 10)$ give M1A1 for 0.84 or better, otherwise M0A0										
		ii		A1	cao											
						Examiner's Comments										
					Only about 25% of candidates gained marks in this part - most either misread or misinterpreted the question and calculated the probability of needing exactly 10 attempts. Some candidates read the question correctly but spent time calculating all the probabilities from 1 to 10 and summing them, usually successfully. Very few used the method stated in the mark scheme.											
			Total	5												
2		i	<table><tr><td><i>r</i></td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>P(<i>X</i> = <i>r</i>)</td><td>3<i>k</i></td><td>8<i>k</i></td><td>15<i>k</i></td><td>24<i>k</i></td></tr></table>	<i>r</i>	2	3	4	5	P(<i>X</i> = <i>r</i>)	3 <i>k</i>	8 <i>k</i>	15 <i>k</i>	24 <i>k</i>	B1	For correct table (ito <i>k</i> or correct probabilities 0.06, 0.16, 0.30, 0.48)	
<i>r</i>	2	3	4	5												
P(<i>X</i> = <i>r</i>)	3 <i>k</i>	8 <i>k</i>	15 <i>k</i>	24 <i>k</i>												

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$3k + 8k + 15k + 24k = 1$	M1		For their four multiples of k added and $= 1$.
		i	$k = 0.02$	A1	or $k = 1/50$ (with or without working) Examiner's Comments Nearly all candidates were able to calculate the correct coefficients of k and sum the terms to get $50k$. Work was generally neatly presented and well structured. Only a very few candidates failed to get the correct answer of $1/50$. A small number of candidates did not show the probabilities in a table, thus losing 1 mark.	Allow M1A1 even if done in part (ii) – link part (ii) to part (i)
		ii	$E(X) = (2 \times 0.06) + (3 \times 0.16) + (4 \times 0.30) + (5 \times 0.48) = 4.2$	M1	For $\sum rp$ (at least 3 terms correct) Provided 4 reasonable probabilities seen.	If probs wrong but sum = 1 allow full marks here. If sum $\neq$ allow max M1A0M1M0A0 (provided all probabilities between 0 and 1)
		ii	or $21/5$	A1	cao	Or ito k NB $E(X) = 210k$, $E(X^2) = 924k$ gets M1A0M1M0A0. $E(X) = 210k$, $\text{Var}(X) = 924k - (210k)^2$ gets M1A0M1M1A0.
		ii	$E(X^2) = (4 \times 0.06) + (9 \times 0.16) + (16 \times 0.30) + (25 \times 0.48) = 18.48$	M1	For $\sum r^2p$ (at least 3 terms correct)	
		ii	$\text{Var}(X) = 18.48 - 4.2^2$	M1	dep for – their $E(X)^2$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	= 0.84 = 21/25	A1	FT their $E(X)$ provided $\text{Var}(X) > 0$ (and of course $E(X^2)$ is correct) Examiner's Comments Once again, a substantial majority of candidates scored full marks. The most common errors were just calculating $E(X^2)$ believing that to be the variance, finding $E(X^2) - E(X)$ or dividing their answers by 4 or some other factor. Those whose probabilities did not sum to 1 were only able to gained two marks out of the 5 available.	Use of $E(X - \mu)^2$ gets M1 for attempt at $(x - \mu)^2$ should see $(-2.2)^2$, $(-1.2)^2$, $(-0.2)^2$, 0.8^2, (if $E(X)$ wrong FT their $E(X)$) (all 4 correct for M1), then M1 for $\sum p(x - \mu)^2$ (at least 3 terms correct with their probabilities) Division by 4 or other spurious value at end gives max M1A1M1M1A0, or M1A0M1M1A0 if $E(X)$ also divided by 4. Unsupported correct answers get 5 marks
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3		i	$X \sim B(50, 0.1)$	M1	For $0.1^5 \times 0.9^{45}$	
		i	$P(5 \text{ underweight}) = \binom{50}{5} \times 0.1^5 \times 0.9^{45} = 0.1849$	M1	For $\binom{50}{5} \times p^5 \times q^{45}$	With $p + q = 1$ Also for $2118760 \times 8.73 \times 10^{-8}$
		i		A1	CAO <u>Examiner's Comments</u> This question was very well answered, with most candidates scoring all 3 marks. However, a few candidates seemed to have no idea about the binomial distribution.	Allow 0.185 or better NB 0.18 gets A0
		ii	$X \sim B(20, 0.1)$			
		ii	$P(X \geq 1) = 1 - P(X = 0)$	M1	For 0.1216	Allow M1 for 0.9^{20}
		ii	$= 1 - 0.1216 = 0.8784$	A1	CAO <u>Examiner's Comments</u> Again another well answered question, although occasionally candidates did not read the question carefully and continued to use $n = 50$ in their calculation.	Allow 0.878 or better See tables at the website http://www.mei.org.uk/files/pdf/formula_book_mf2.pdf
		iii	$E(X) = 48 \times 0.8784 = 42.16$ (= 42.2)	M1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		A1	FT their probability from part (ii) Examiner's Comments Full marks were available here for a correct follow through from part (ii), so many candidates managed to recover from an incorrect answer. However a large proportion of candidates rounded their answer to the nearest whole number, thus losing a mark. Others over-specified their final answer, again losing a mark. Other common errors were to use $p = 0.1$, rather than their answer to part (ii), or to use $n = 48 \times 20$.	If any indication of rounding to 42 or 43 or to another integer on FT allow M1A0 SC1 for $48 \times$ their p giving an integer answer. NB 0.6083 in (ii) leads to 29.20 NOTE RE OVER-SPECIFICATION OF ANSWERS If answers are grossly over-specified, deduct the final answer mark in every case. Probabilities should also be rounded to a sensible degree of accuracy. In general final non probability answers should not be given to more than 4 significant figures. Allow probabilities given to 5 sig fig. PLEASE HIGHLIGHT ANY OVER-SPECIFICATION Please note that there are no G or E marks in scoris, so use B instead
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		i	$3 \times \frac{5}{10} \times \frac{4}{9} \times \frac{5}{8} = \frac{300}{720} = \frac{5}{12} = (0.4167)$	M1	For $5/10 \times 4/9$	Correct working but then multiplied or divided by some factor scores M1M1M0A0
		i		M1	For $\times 5/8$	Zero for binomial Allow M2 for equivalent triple such as $\frac{5}{10} \times \frac{5}{9} \times \frac{4}{8}$
		i		M1	For $3 \times$ triple product	Or 3 separate equal triplets added
		i		A1	CAO (Fully simplified)	Answer must be a fraction
		i	Or $\frac{\begin{pmatrix} 5 \\ 2 \end{pmatrix} \times \begin{pmatrix} 5 \\ 1 \end{pmatrix}}{\begin{pmatrix} 10 \\ 3 \end{pmatrix}} = \frac{10 \times 5}{120} = \frac{5}{12}$	M1*	For $\begin{pmatrix} 5 \\ 2 \end{pmatrix} \times \begin{pmatrix} 5 \\ 1 \end{pmatrix}$	Seen
		i		M1*	For $\begin{pmatrix} 10 \\ 3 \end{pmatrix}$	Seen
		i		M1*dep	For whole fraction	Correct working but then multiplied or divided by some factor scores M1M1M0A0

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i		A1	CAO (Fully simplified) <u>Examiner's Comments</u> Candidates using the nC_r method tended to be more successful, as when using the product of 3 fractions method many did not realise that they needed to multiply the final product by 3. A small minority of candidates did not follow instructions and either left a fraction in unsimplified form (usually 15/36) or gave the answer as a decimal.	
		ii	$4 \times \frac{7}{12} \times \left(\frac{5}{12}\right)^3 + \left(\frac{5}{12}\right)^4$	M1FT	For first probability	Allow 4C_3
		ii		M1FT	For $(5/12)^4$	
		ii		M1FT	For sum of both correct probabilities	Provided sum <1

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= 0.169 + 0.030 \times 0.199$ Or = $\frac{875}{5184} + \frac{625}{20736} = \frac{1375}{6912}$	A1	CAO Do not allow 0.2, unless fuller answer seen first <u>Examiner's Comments</u> Most candidates made a reasonable start in this part, using their answer from part (i). However, many only calculated one probability, or missed the coefficient of 4 when calculating the probability of 3 evenings, not realising this was a binomial situation. Some candidates calculated the probability of 3, rather than <i>at least</i> 3, and thus only gained 1 mark. A small minority of candidates used statistical functions on graphical calculators to just write down an answer – this was a risky strategy, as a slip in copying the answer was heavily penalised, since no method was shown.	Alternative for 1- (P(0) + P(1) + P(2)) allow M1FT for two 'correct' probs, M1 for sum of three 'correct', M1 for 1 – answer, A1 CAO NOTE RE OVER-SPECIFICATION OF ANSWERS If answers are grossly over-specified, deduct the final answer mark in every case. Probabilities should also be rounded to a sensible degree of accuracy. In general final non probability answers should not be given to more than 4 significant figures. Allow probabilities given to 5 sig fig. PLEASE HIGHLIGHT ANY OVER-SPECIFICATION Please note that there are no G or E marks in scoris, so use B instead
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
5		i	$\binom{25}{15}$ $= 3268760$	M1	Examiner's Comments This was very well-answered.	Accept ${}^{25}C_{15}$ or ${}^{25}P_{(15!10!)}$ or equivalent for M1 No marks for permutations
		i		A1		Exact answer required
		ii	$\binom{13}{8} \times \binom{12}{7} = 1287 \times 792$ $= 1019304$	M1	For product of both correct combinations	No marks for permutations
		ii		A1	CAO Examiner's Comments This was again usually answered well although a fairly common error was to add the two combinations rather than to multiply them.	Exact answer required
		iii	1019304/3268760	M1	For their (ii) divided by their (i)	Allow 0.31 with working SC1 for $\binom{15}{8} \times \left(\frac{13}{25}\right)^8 \times \left(\frac{12}{25}\right)^7$ Allow $\binom{15}{8}$ or $\binom{15}{7}$
		iii	= 0.312 (0.311832) Allow fully simplified fraction 11583/37145	A1FT		
		iii	OR			
		iii	$\binom{15}{8} \times \frac{13}{25} \times \frac{12}{24} \times \frac{11}{23} \times \frac{10}{22} \times \frac{9}{21} \times \frac{8}{20} \times \frac{7}{19}$ $\times \frac{6}{18} \times \frac{12}{17} \times \frac{11}{16} \times \frac{10}{15} \times \frac{9}{14} \times \frac{8}{13} \times \frac{7}{12} \times \frac{6}{11}$	M1	For product of fractions with coefficient	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	= 0.312	A1	Examiner's Comments This was another well-answered question with even those who had added in part (ii) still usually scoring both marks on follow through. Candidates who tried to use a probability method (instead of simply dividing their answer to part (i) by their answer to part (ii)) were rarely successful, and even if they did have the correct product of 15 probabilities, they rarely multiplied this by any, let alone the correct combination.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6		i	$P(X < 450) = P\left(Z < \frac{450 - 435}{30}\right)$	M1	For standardising. M0 if 'continuity correction' applied	
		i	$= P(Z < 0.5) = \Phi(0.5)$	M1	For correct structure	
		i	$= 0.6915$	A1	CAO Allow 0.692	
					Examiner's Comments Well answered, though inappropriate "continuity corrections" were seen on occasion.	
		i	$P(400 < X < 450)$ $= P\left(\frac{400 - 435}{30} < Z < \frac{450 - 435}{30}\right)$ $= P(-1.1667 < X < 0.5)$ $= \Phi(0.5) - \Phi(-1.1667)$	M1	For correct structure	
		i	$= 0.6915 - 0.1216$	B1	For use of difference column to obtain 0.8784, 0.8783, 0.1216 or 0.1217. Condone 0.8782 or 0.1218	
		i	$= 0.5699$	A1	FT "their 0.6915" – 0.1216 (or 0.1217)	
					Examiner's Comments Well answered, though arithmetic errors were quite common. In several cases, –1.667 was used rather than –1.1667 often as a result of candidates misreading their own figures. A few candidates lost accuracy by prematurely rounding their z-value before using the Normal tables.	
		ii	P(all 5 between 400 and 450)			
		ii	$= 0.5699^5$	M1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	= 0.0601	A1	FT Allow 0.060	
					Examiner's Comments Very well answered. Most candidates scored both marks.	
		iii	$P(Y < 350) = 0.2, P(Y > 390) = 0.1$ $P(Z < \frac{350 - \mu}{\sigma}) = 0.2$ $\Phi^{-1}(0.2) = -0.8416$			
		iii	$\frac{350 - \mu}{\sigma} = -0.8416$	M1	For equation as seen or equivalent with their -ive z value	If 'continuity corrections' applied allow M marks but do not award final A marks
			$P(Z > \frac{390 - \mu}{\sigma}) = 0.1$			
		iii	$\Phi^{-1}(0.9) = 1.282$	B1	For 1.282 or -0.8416	
		iii	$\frac{390 - \mu}{\sigma} = 1.282$ $350 = \mu - 0.8416 \sigma$ $390 = \mu + 1.282 \sigma$ $2.1236 \sigma = 40$	M1	For equation as seen or equivalent with their +ive z value	
		iii	$\sigma = 18.84$	A1	Allow 18.8	Answers to max 2 d.p.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\mu = 350 + (0.8416 \times 18.84)$ $= 365.85$	A1	Allow 365.86, 366, 365.9 Examiner's Comments On the whole, this was well answered. Many candidates provided clear, accurate methods leading to correct final answers. Some candidates started out with one of the required equations containing a sign error which was not picked up, even when the error led to a negative value for σ . Most candidates identified the correct z-values. In the poorest answers, continuity corrections were attempted and z-values were changed to absurd values, such as "1 – 0.8416", before substitution into equations. Over-specification of final answers was seen, on occasion, here.	
		iv	$\Phi^{-1}(0.975) = 1.96$	B1	For using a suitable pair of z values e.g. ± 1.96	Accept any correct values of a and b .
		iv	$a = 365.85 - (1.96 \times 18.84)$	M1	For either equation provided that a suitable pair of z-values is used. e.g. +2.326 and –1.751	
		iv	$= 328.9$	A1	FT their μ and σ to 2 d.p. (A0 if 'continuity correction' used) Examiner's Comments Though one of the more challenging parts, many candidates scored full marks here. A variety of correct, "non-symmetrical" solutions were seen though most opted to use z-values of ± 1.96 .	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$b = 365.85 + (1.96 \times 18.84)$ $= 402.8$	A1	FT their μ and σ to 2 d.p. (A0 if 'continuity correction' used)	
			Total	17		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
7		i	$P(Y = 76) = P\left(\frac{75.5 - 76}{12} \leq Z \leq \frac{76.5 - 76}{12}\right)$ $= P(-0.04166... < Z < 0.04166...)$ $= \Phi(0.04166...) - (1 - \Phi(0.04166...))$ $= 2 \times \Phi(0.04166...) - 1$ $= 2 \times 0.5167 - 1$ $= 0.0334$	B1	For one correct continuity correction used	
		i		M1	For standardizing	
		i		M1	For correctly structured probability calculation.	
		i		A1	CAO inc use of diff tables. Allow 0.0330 – 0.0340 www. <u>Examiner's Comments</u> Most candidates obtained a correct answer. A small but significant number did not use one or both of the continuity corrections. Most used the difference column of the Standard Normal table correctly to provide suitably accurate answers. A relatively small number struggled with the structure of the calculation.	
		ii	$P(Y \geq 80) = P\left(Z \geq \frac{79.5 - 76}{12}\right)$ $= P(Z > 0.2917) = 1 - \Phi(0.2917)$ $= 1 - 0.6148 = 0.3852 = 0.385 \text{ to 3 sig fig}$	B1	For correct cc used	
		ii		M1	For correct structure	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	CAO do not allow 0.386 <u>Examiner's Comments</u> Many candidates answered correctly, but a common mistake was to omit or to provide an incorrect continuity correction. A relatively small number did not use the difference column correctly.	
		iii	$3 \times 0.3852 \times 0.6148^2 = 0.4368$	M1	$3 \times \text{their } p \times (1 - \text{their } p)^2$	
		iii		A1	FT their p . Allow 2sf if working seen. <u>Examiner's Comments</u> Most candidates knew and applied the method correctly but many were dependent on the FT to gain the 2 marks. A small number omitted the x3 from their binomial calculation.	
		iv	EITHER: $P(\text{Score} \geq k) = 0.1$			
		iv	$\Phi^{-1}(0.9) = 1.282$	B1	For 1.282	
		iv		M1	Allow $k - 0.5$ used for k . Positive z used.	
		iv	$k = 76 + (1.282 \times 12) = 91.38$ or $k = 76 + 0.5 + (1.282 \times 12) = 91.88$	A1	For 91.38 or 91.88	
		iv	$91.38 > 90.5$ or $91.88 > 91$	M1	Relevant comparison (e.g. diagram)	www
		iv	so lowest reported mark = 92	A1		
		iv	OR Trial and improvement method	M1	M1 for attempt to find $P(\text{Mark} \geq \text{integer})$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$P(\text{Mark} \geq 91) = P(\text{Score} \geq 90.5) = 0.1135$	A1	A1 for 0.1135	
		iv	$P(\text{Mark} \geq 92) = P(\text{Score} \geq 91.5) = 0.0982$			
		iv	$P(\text{Mark} \geq 91) > 10\%$ and $P(\text{Mark} \geq 92) < 10\%$	M1	M1 for comparisons	www
					<u>Examiner's Comments</u> 1.282 was identified by the majority of candidates who went on to set up a correct equation and arrive at 91.38 or 91.88. Many of these gave 92% as the final answer but many others gave 91%. Others rearranged incorrectly and arrived at 60.6 for the first calculation. Few candidates demonstrated a proper understanding of the requirement of this question.	
		iv	so lowest reported mark = 92	A1		
		v	$P(Y \leq 50) = 0.2$			
		v		B1	For 50.5 used	
		v		B1	For -0.8416. Condone - 0.842	
					Condone 0.8416 if numerator reversed.	
		v		M1	For structure.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	$\mu = 50.5 + (12 \times 0.8416) = 60.6$	A1	CAO <u>Examiner's Comments</u> In this part, the continuity correction was omitted, or an incorrect value was used, by many candidates. Many used +0.8416 leading to 60.0992 which was a common answer. The issue of over-specification was most apparent in this part of the question.	
			Total	18		

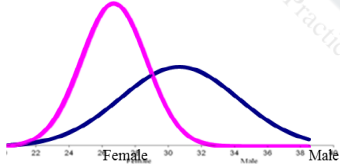
Question			Answer/Indicative content	Marks	Part marks and guidance	
8		i	$P(X > 135) = P\left(Z > \frac{135-130.5}{\sqrt{11.84}}\right)$	M1	For standardising. Penalise use of “continuity corrections”	
		i	$= P(Z > 1.308) = 1 - \Phi(1.308) = 1 - 0.9045$	M1	For correct structure i.e. finding the area to the right of their z	
		i	$= 0.0955$ (3s.f.)	A1	CAO inc use of diff tables Allow 0.0954 and 0.0956 If numerator reversed, give BOD only if $P(Z < -1.308)$ is used Examiner's Comments Was found to be easy, though some candidates confused the exact and approximate values in calculating the relative error, thereby losing a mark.	
		ii	From tables $\Phi^{-1}(0.99) = 2.326$	B1	± 2.326 or (better) seen, not ± 2.33	
		ii	$\frac{k-130.5}{\sqrt{11.84}} = 2.326$ oe	M1	For sensible equation in k with their z value. Note that use of $z = 0.8389$ from $\Phi(0.99)$ gets BOM0A0, as 0.8389 is clearly a probability . Allow use of -2.326 (or their negative z) with numerator reversed. Condone use of $\sigma = 11.84$ if also used in part (i). Condone use of “ $k \pm 0.5$ ” for k in equation. 0/3 for trial and improvement	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$k = 130.5 + (2.326 \times \sqrt{11.84}) = 138.50$	A1	CAO Allow 138.504. Accept 138.5 Do not accept final answers of 139 or 138. Examiner's Comments Was found hard. Very few candidates spotted that squaring and cubing gave double and triple the relative errors.	
		iii	$P(\text{Wing length} = 131) = P\left(\frac{130.5 - 130.5}{\sqrt{11.84}} \leq Z \leq \frac{131.5 - 130.5}{\sqrt{11.84}}\right)$	B1	For both limits correct, soi. e.g. use of 0.5 in probability calculation implies correct lower limit.	
		iii	$= P(0 < Z < 0.2906)$ $= \Phi(0.2906) - \Phi(0)$ $= 0.6143 - 0.5$	M1	For correct structure using their standardised values. i.e. Finding the area between their z values found using $\mu = 130.5$	
		iii	$= 0.1143$	A1	Condone use of $\sigma = 11.84$ if also used in part (i) or part (ii). CAO inc use of diff tables Allow 0.1145 Allow 0.114 www	
		iv	$H_0: \mu = 130.5$ $H_1: \mu > 130.5$	B1	For both correct Hypotheses in words must refer to population. Do not allow other symbols unless clearly defined as population mean	
		iv	Where μ denotes the mean wing length (in the population) (of Scandinavian male blackbirds).	B1	For definition of μ in context. Do not allow "sample mean wing length" or "mean wing length of English blackbirds"	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	Test statistic = $\frac{132.4 - 130.5}{\sqrt{11.84} / \sqrt{20}} = \frac{1.90}{0.7694}$	M1*	must include $\sqrt{20}$ Condone use of $\sigma = 11.84$ if also used in part (i), part (ii) or part (iii). Condone numerator reversed for max M1*A0B1M0depA0A0 (max 4/8)	
		iv	= 2.469	A1	Allow 2.47	
		iv	Upper 5% level 1 tailed critical value of $z = 1.645$	B1	For 1.645. Must be positive. B0 if -1.645 seen. No further A marks from here if wrong.	
		iv	$2.469 > 1.645$	M1dep*	For sensible comparison leading to a conclusion.	
		iv	The result is significant. There is sufficient evidence to reject H_0	A1	For correct conclusion. e.g. for "significant" oe FT only candidate's test statistic if $cv = 1.645$	
		iv	There is sufficient evidence to suggest that the mean wing length (of this population of birds) is greater (than 130.5 mm).	A1	For non-assertive conclusion in context, consistent with their result Condone use of "average" for "mean" FT only candidate's test statistic if $cv = 1.645$	
		v	With a 10% significance level rather than a 5% significance level, Advantage: One is less likely to accept the null hypothesis when it is false.	E1	Accept equivalent wording.	
		v	Disadvantage: One is more likely to reject the null hypothesis when it is true.	E1	Note — Unless stated otherwise, assume the first comment relates to an advantage and the second comment relates to a disadvantage.	
			Total	19		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	(A) $P(X < 30)$			
		i	$= P\left(Z < \frac{30 - 30.7}{3.5}\right)$	M1	For standardising	
		i	$P(Z < -0.20)$			
		i	$= \Phi(-0.20)$			
		i	$= 1 - \Phi(0.20)$ $= (1 - 0.5793)$	M1	For correct structure	1 - Φ (positive z)
		i	$= 0.4207$	A1	CAO	Allow 0.421 www
					Examiner's Comments	
					The majority of candidates correctly standardised the given value then found the correct probability. Some candidates made good use of diagrams to indicate their intent.	
		i	(B) $P(25 < X < 35)$			
		i	$= P\left(\frac{25 - 30.7}{3.5} < Z < \frac{35 - 30.7}{3.5}\right)$	M1	Correctly standardising both.	Penalise erroneous continuity corrections and wrong sd. Condone both numerators reversed.
		i	$= P(-1.629 < X < 1.229)$ $= \Phi(1.229) - \Phi(-1.629)$			
		i	$= 0.8904 - (1 - 0.9483) = 0.8904 - 0.0517$	M1	For correct structure	$\Phi(1.23) - \Phi(-1.63)$ leads to $0.8907 - 0.0516 = 0.8391$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	= 0.8387	A1	Use of differences column required <u>Examiner's Comments</u> For a question that had potential for many errors, either in standardising, approximating too early, or looking up values incorrectly, this question was answered well. This part was in general done rather better than part (iA).	Only allow 0.839 if 0.8387 is seen.
		ii	P(all 5 weigh at least 30kg)			
		ii	= 0.5793 ⁵	M1	Allow FT (1 – their (i)(A)) ⁵ or [their P(X ≥ 30)] ⁵	
		ii	= 0.0652	A1	FT only (1 – their (i)(A)) ⁵ <u>Examiner's Comments</u> The binomial situation, within a question fundamentally on the Normal distribution, was dealt with well by many candidates. The errors that did occur were often about forgetting that (iA) found p(X < 30) whereas this part asked for all five dogs to weigh <i>more than</i> 30 kg.	Allow 0.06524, allow 0.065 www
		iii	P(weight > 30) = 0.05 $P(Z > \frac{30 - 26.8}{\sigma}) = 0.05$			
		iii	$\Phi^{-1}(0.95) = 1.645$	B1	For 1.645. B0 for 1 - 1.645 or 0.1645	NOTE use of –1.645 allowed only if numerator
		iii	$\frac{30 - 26.8}{\sigma} = 1.645$	M1*	For equation as seen or equivalent, with their z > 1.	reversed. Condone use of spurious c.c. if already penalised in parts (i)(A) or (i)(B). See additional guidance notes. Allow $\sigma = 1.95$ www

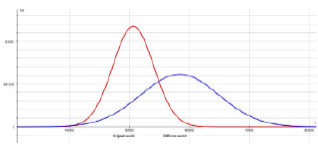
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\sigma = \frac{30 - 26.8}{1.645} = 1.945 \text{ kg}$	M1dep*	Rearranging for σ	
		iii		A1	CAO <u>Examiner's Comments</u> Few candidates used the wrong tail for this calculation, most correctly identifying that +1.645 was the appropriate value to use as 5% weighed <i>more than</i> 30 kg.	<u>Additional Notes for (iii)</u> M1* is for forming a suitable equation using their z-value but it must be reasonably clear that the value used is a z-value – for example we do not allow 0.05 or 0.95 to be treated as z-values here. The M1dep* can be awarded if the candidate correctly rearranges their equation to find σ . Hence, use of an incorrect z-value could earn max BOM1*M1dep*A0. However, if it is clear that the z-value is from the wrong tail (e.g. -1.645 used in place of +1.645) then award 0/4. In cases where -1.645 is used and the numerator of the equation is reversed allow full credit and annotate with BOD.
		iv		G1	For two Normal shapes including attempt at asymptotic behaviour with horizontal axis at each of the four ends	Penalise clear asymmetry
		iv		G1	For means, shown explicitly or by scale on a single diagram	If shown explicitly, the positions must be consistent with horizontal scale if present.
		iv		G1	For lower max height for Male	If not labelled, assume the larger mean represents Male

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv		G1	For visibly greater width for Male <u>Examiner's Comments</u> This part of the question tested most candidates. Thus many found dealing with the correct shape and the various relative constraints on the curves difficult. The mark scheme will help with interpretation of the requirements.	If not labelled, assume the larger mean represents Male
			Total	16		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		i	$P(50000 < X < 55000) =$ $P\left(Z \geq \frac{750-751.4}{2.5}\right) \left(\frac{50000-50600}{3400} < Z < \frac{55000-50600}{3400}\right)$	M1	For standardising both. SOI. Penalise erroneous continuity corrections and wrong sd. Condone numerator(s) reversed.	
		i	$= P(-0.176 < Z < 1.294) =$ $\Phi(1.294) - (1 - \Phi(0.176)) =$ $0.9022 - 1 + 0.5699$	M1	For correct structure $\Phi(\text{positive } z) - \Phi(\text{negative } z)$	
		i	$= 0.4721$	A1	CAO including use of difference tables (Answer from calculator 0.4722 and from tables interpolated 0.4723)	
					Examiner's Comments Well answered. Errors caused by lack of accuracy reading Normal tables were seen fairly regularly. Most candidates used the correct probability structure with their z values.	
		ii	$\left(Z > \frac{45000 - 50600}{3400} \right)^{P\left(Z \geq \frac{750-751.4}{2.5}\right)} =$ $P(Z > -1.647)$	B1*	For -1.647 or $-\Phi^{-1}(0.95) = -1.645$ or 1.647 seen with $P(X < 56200)$ or numerator reversed	
		ii	$= \Phi(1.647) = 0.9502$	B1*	For 0.9502 or 45007 or 0.0498, or B1 for -1.645 if B1 for -1.647 already awarded.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	0.9502 > 95% so agree with claim	depE1*	For comparison seen e.g. $-1.647 < -1.645$ or $0.0498 < 0.05$ or $1.647 > 1.645$ or 95% last longer than 45007 hours, and correct conclusion. Dependent on B1, B1 awarded Examiner's Comments This was well done on the whole though many candidates did not provide the required comparison to justify their conclusion. In most cases the working provided was clear – diagrams were helpful to examiners in conveying the candidates' intentions – often more successfully than their wording.	
		iii	From tables $\Phi^{-1}(0.999) = 3.09$	B1	±3.09 seen	
		iii	$\frac{h - 50600}{3400} = -3.09$	M1	For equation as seen with their negative z-value	
		iii	$k = 50600 - (3.09 \times 3400) = 40100$ www	A1	CAO Allow 40094, 40090 Examiner's Comments Many correctly identified the z value of -3.09 and went on to find the appropriate value for h, rounded to a suitable level of accuracy.	
		iv	$P(Y < 60000) = 0.6 \square P$ $\left(Z < \frac{60000 - \mu}{\sigma} \right) = 0.6$	B1	For ±0.2533 or ±1.282 seen	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$\square \frac{60000 - \mu}{0.2533 \sigma} = \Phi^{-1}(0.6) =$	M1	For an equation into μ , σ , z and y formed. NB using $z = \pm 0.2533$ with $y = 60000$ or ± 1.282 with $y = 50\,000$	
		iv	$\square 60000 = \mu + 0.2533 \sigma$			
		iv	$P(Y > 50000) = 0.9 \quad \square P$ $\left(Z > \frac{50000 - \mu}{\sigma} \right)$ $= 0.9$			
		iv	$\square \frac{50000 - \mu}{\sigma} = \Phi^{-1}(0.1) = -1.282$	A1		
		iv	$\square 50000 = \mu - 1.282 \sigma$ $1.5353 \sigma = 10000$			
		iv	$\sigma = 6513$	A1	CAO Allow 6510, 6515	
		iv	$\square \mu = 50000 + (1.282 \times 6513) = 58350$	A1	CAO Allow 58400	
					Examiner's Comments There were some pleasing attempts at this question, marred only by an inappropriate degree of accuracy for the final answers. It was good to see that once the initial equations had been established with the correct z values, many could still solve the simultaneous equations. A few candidates failed to identify, for the given probabilities, the z values needed for the simultaneous equations.	

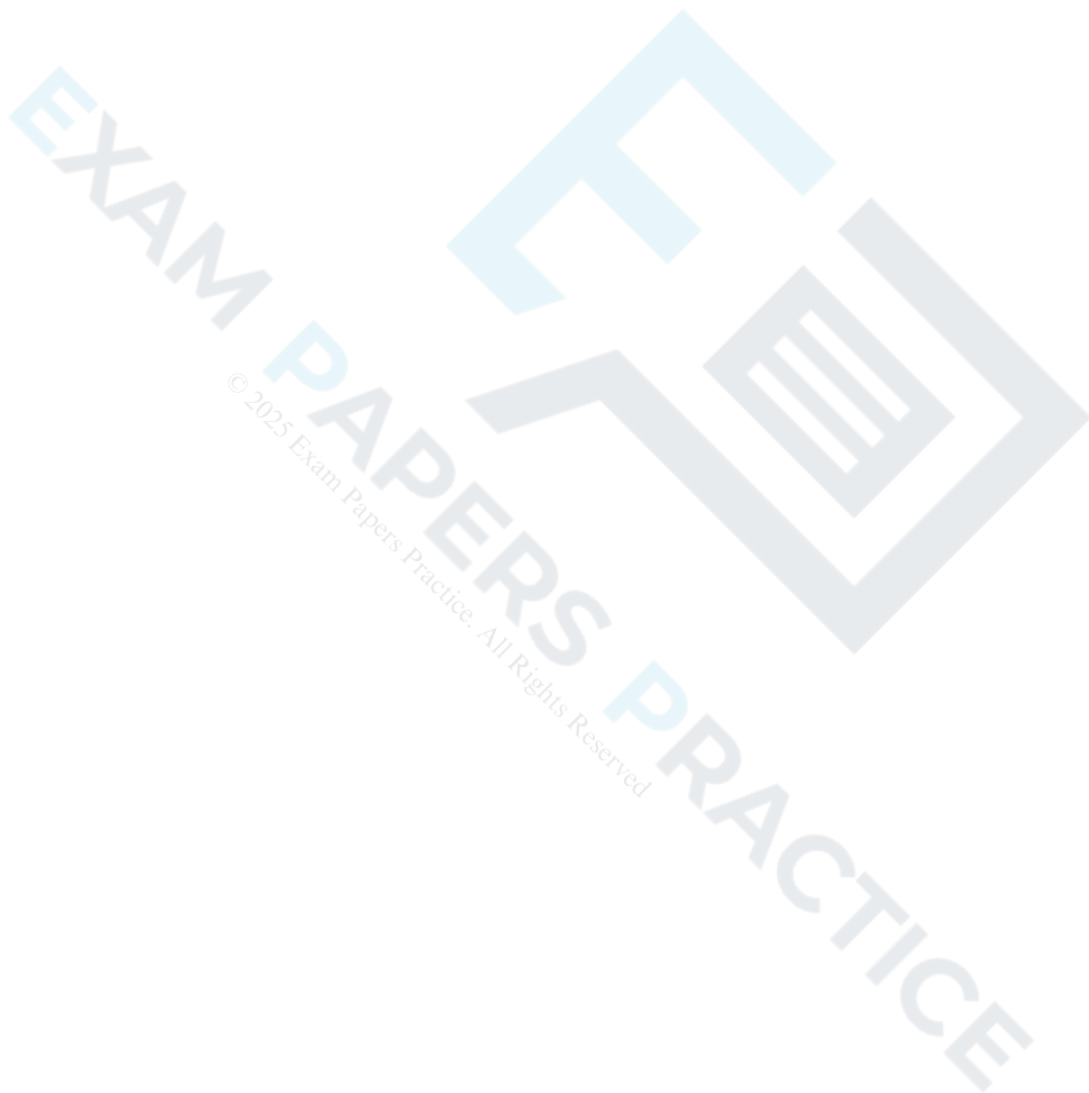
Question			Answer/Indicative content	Marks	Part marks and guidance	
		v		G1	For two Normal shapes including attempt at asymptotic behaviour with horizontal axis at each of the four ends. Penalise clear asymmetry.	
		v		G1	For means, shown explicitly or by scale on a single diagram. If shown explicitly, the positions must be consistent with horizontal scale if present. FT part (iv).	
		v		G1	For greater width (variance) for Different model. FT part (iv).	
		v		G1	For lower max height for Different model. FT part (iv) If not labelled assume the larger mean represents Different model. FT part (iv).	
					Examiner's Comments This was answered well, though many candidates could have made a greater effort to include symmetry in their sketches and to pay more attention to the asymptotic nature. Spurious labelling of axes was seen but only rarely.	
			Total	18		
11			Binomial(20, 0.08) $P(2 \text{ blue}) = 0.27[11]$	M1(AO3.3) A1(AO1.1) [2]	BC	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12		a	i Mean = 17	B1(AO3.4)		
		a	ii Either Points of inflection are approx. 3 above and below mean so SD = approx. 3 Or Limits are approx. 9 above and below mean so SD = $9 \div 3 = 3$	E1(AO2.4) E1(AO2.4) [2]	AG AG	
		b	Mean in Fahrenheit = $1.8 \times 17 + 32 = 62.6$ SD in Fahrenheit = $1.8 \times 3 = 5.4$	B1(AO1.1) B1(AO1.1) [2]	FT their mean	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance		
13		a	Estimated number $= 4 + \frac{16}{3} = 9\frac{1}{3}$ $\frac{9\frac{1}{3}}{80} = 0.1166\dots$ so proportion is approximately 0.117	M1(AO3.1b) A1(AO1.1) [2]		for attempt at interpolation	
		b	E.g. Midpoints Mean Standard deviation = 3.4	M1(AO1.1) A1(AO1.1) A1(AO1.1) [3]	evidence of valid method for estimation BC Mean in the range 169-171 BC SD in the range 3-3.5		
		c	The histogram e.g. seems to have a rough bell shape e.g. is symmetrical (around the estimated mean) e.g. appears to have all data within 3 s.d. of the mean so this does support the manager's belief	B1(AO3.5a) B1(AO3.5a) [2]	for one reason for at least two reasons and 'supports belief'		
		d	i P(Lifetime > 174) for N(170, 3.4 ²) 0.1197	M1(AO3.4) A1(AO1.1)	oe BC FT their mean and standard deviation		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	ii Answer is very similar to estimate in part (a)	B1(AO3.5a)		
				[3]		
		e	Either Test statistic $= \frac{207.3 - 210}{3.4 / \sqrt{8}} = -2.246$ Lower 5% level 1 tailed critical value of $z = -1.645$ $-2.246 < -1.645$ so significant or under $H_0, \bar{X} \sim N\left(210, \frac{3.4^2}{8}\right)$ $P(\bar{X} \leq 207.3) = 0.01235$ $0.01235 < 0.05$ so significant There is sufficient evidence to reject H_0 There is sufficient evidence to conclude that the mean lifetime is less than 210 minutes.	M1(AO3.4) A1(AO1.1) B1(AO1.1) M1(AO3.4) A1(AO1.1) B1(AO1.1) A1(AO2.2b) E1(AO2.4) [5]	Must include $\sqrt{8}$ For comparison leading to correct conclusion BC	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	15	


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Question			Answer/Indicative content	Marks	Part marks and guidance		
14		a	Comment about shape of distribution for first graph Comment about shape of distribution for second graph	B1(AO2. 2b) B1(AO2. 2b) [2]	Comments can be combined e.g Both distributions negatively skewed gets both marks e.g. 1974 distribution has greater spread than 2014 gets both marks	If zero scored, SC1 for “The 2014 distribution is shifted to the right of the 1974 distribution” oe	
		b	i Life expectancy went down [between 1974 and 2014] in [at least] one country	E1(AO2. 2a) [1]	NOT increase in life expectancy is negative		
		b	ii The box plot is not symmetrical.	B1(AO3. 5b) [1]			
		b	iii Not appropriate with reason	E1(AO2. 4) [1]	e.g. [some] values of life expectancy are estimates The values of life expectancy are not available to this level of accuracy		
		b	iv Comment about life expectancy at birth data for countries and not individual people	B1(AO2. 4) [1]			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	Use of $Q3 + 1.5 \times (Q3 - Q1)$ $15.873 + 1.5(8.9154) = 29.2461$ (years) The maximum value is an outlier as $30.742 > 29.2461$.	M1(AO1.2) M1(AO1.1) A1(AO1.1) [3]		
		d	<i>i</i> approx $60.8 - 37.5 = 23.3$ (years)	M1(AO3.1b)	Attempt to estimate change in life expectancy at birth soi.	
		d	<i>ii</i> Change in life expectancy for Sweden approx $81.9 - 72.5 = 9.4$ (years)	A1(AO1.1 1.1)	FT 'their 37.5 between 35 - 40'	
		d	<i>iii</i> E.g. Countries with a lower life expectancy in 1974 have greater opportunity to increase life expectancy in 2014.	A1 E1 (AO3.2a) [4]	FT 'their 72.5 between 70 - 75' OR Countries with a higher life expectancy in 1974 have less opportunity to increase life expectancy in 2014.	
		e	<i>i</i> $30.98 + 0.67 \times 37.4 = 56.0$ (years)	M1(AO3.4) A1(AO1.1) [2]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		e	ii E.g. Large amount of scatter at the lower values [and South Sudan is 37.4]. E.g. Not having the data value could indicate that there are problems in the country which could mean it does not follow the pattern for other countries	E1(AO3.5b) E1(AO3.5b) [2]	E1 Reason inferred from Fig 16.4 E1 For knowing why data may be missing	
		f	Correct method Clearly explained 6	M1(AO3.1b) E1(AO2.4) A1(AO1.1) [3]	e.g. draw “y = x” on graph e.g. The value on the vertical axis must be lower than the one on the horizontal axis FT their correct method	
			Total	20		

Question			Answer/Indicative content	Marks	Part marks and guidance		
15		a	(A) $X \sim B(30, 0.92)$, $P(X = 28)$ $= 0.2696$	B1(AO3.3) B1(AO1.1) [2]	BC		
		a	(B) $P(X > 27) = 1 - 0.4346$ oe $= 0.5654$	M1(AO1.1) A1(AO1.1) [2]	OR for sum of at least two correct probabilities from $0.2696 + {}^{30}C_{29} \times 0.92^{29} \times 0.08^1 + 0.92^{30}$ BC		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	Let p = probability that a train arrives on time $H_0: p = 0.92$ $H_1: p < 0.92$ Let $X \sim B(18, 0.92)$ $P(X \leq 13) = 0.0116$ [$> 1\%$] $P(X \leq 12) = 0.0021$ [$< 1\%$] The critical region is $X \leq 12$	B1(AO2.5) B1(AO1.1) M1(AO1.1) M1(AO1.1) A1(AO2.2a) [5]	For definition of p For H_0 and H_1 For probability $P(X \leq \text{any whole number value 1 to 18})$, Both $P(X \leq 13)$ and $P(X \leq 12)$ For correct critical region stated	Allow FT from $H_1: p < 0.92$ OR $H_1: p \neq 0.92$	
			Total	9			

Question			Answer/Indicative content	Marks	Part marks and guidance	
16		a	$k[P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)] = 1$ $k(1 + 1 + 2 + 6) = 1$ so $k = 0.1$	M1(AO1.1) A1(AO3.1a) [2]		
		b	$P(X = 3) = 0.6$	B1(AO1.1) [1]		
		c	$X \sim B(32, 0.6)$ 0.073	M1(AO3.3) A1(AO3.4) [2]	FT their 0.6 BC FT their 0.6 or 0.0728 or 0.07283	
			Total	5		
17		a	Negative skew	B1(AO1.2) [1]		
		b	[They are all less than 0.00005] and the table rounds values [to 4 decimal places]	E1(AO2.4) [1]	Allow equivalent explanation in words	
		c	(18)	B1(AO1.1) [1]		
		d	(0)	1(AO2.2a) [1]		
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance		
18		a	0.4×0.6 or 0.6×0.1 $p(\text{in, out}) + p(\text{out, out})$ 0.3	B1(AO 3.1b) M1(AO 3.1a) A1(AO 1.1) [3]	If B0M0 allow SC1 for correct tree diagram with all possible outcomes shown.		
		b	$1 - p(\text{out every night})$ $1 - 0.6 \times 0.1^6$ $= 0.9999994 \approx 1$ so extremely likely	M1(AO 3.1b) A1(AO 1.1) A1(AO 3.2a) [3]			
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance		
19		a	$\Phi^{-1}(0.08) = -1.4051$ BC $\Phi^{-1}(0.81) = 0.8779$ BC their $-1.4051 = \frac{58-\mu}{\sigma}$ their $0.8779 = \frac{86-\mu}{\sigma}$ obtaining a value for μ or σ from their equations $\sigma = 12.3$ $\mu = 75.2$	B1(AO1.1) B1(AO1.1) M1(AO2.1) M1(AO1.1) M1(AO2.5) A1(AO2.4) A1(AO1.1) [7]	allow 2 or 3 s.f.	NB 12.2647368787 and 75.2328329886	
		b	$P(X < 90) = 0.849$ FT their μ and σ BC which is less than 0.95 so claim not justified	M1(AO3.1a) A1(AO3.2a) [2]	their $12.3 \times 1.645 +$ their $75.2 > 90$ which is greater than 90 so claim not justified		
			Total	9			

Question			Answer/Indicative content	Marks	Part marks and guidance		
20		a	$H_0p = 0.2$ $H_1p < 0.2$ p is the proportion of male adult pilot whales $X \sim B(43, 0.2)$ $P(X \leq 3) = 0.0178$ BC $0.0178 < 0.05$ so result is significant There is sufficient evidence to suggest at the 5% level that the proportion of males in the population of adult pilot whales is less than 20%	B1(AO 1.1) B1(AO 2.5) M1(AO 3.3) B1(AO 3.4) M1(AO 1.1) A1(AO 2.2b) [6]	For both hypotheses Definition of p soi Comparison of their p -value with 0.05		
		b	Sample mean is 3.09 Sample variance is 0.295 $H_0 : \mu = 2.98$ $H_1 : \mu \neq 2.98$ μ is the population mean length of female pilot whales $\bar{X} \sim N(2.98, \frac{0.295}{39})$ $p(\bar{X} \geq 3.09) = 0.1029$ $0.1029 > 0.025$ so result not significant There is insufficient evidence at the 5% level to suggest that the mean length of female pilot whales has changed	B1(AO 1.1) B1(AO 1.1) B1(AO 1.1) B1(AO 2.5) M1(AO 3.3) B1(AO 3.4) M1(AO 1.1) A1(AO 2.2b) [8]	soi For both hypotheses soi Comparison of their p -value with 0.025 BC	NB 3.0948 7179487 and 0.2947 09851552	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	14		
21		a	$0.4a + 0.5a + 0.3a + 0.2a + 0.1a = 1$ so $a = \frac{2}{3}$	M1(AO1.1b) A1(AO1.1b) [2]		
		b	$\frac{2}{3}(0.3 + 0.2 + 0.1)$ 0.4	M1(AO1.1b) A1(AO1.1b) [2]	<i>their</i> $\frac{2}{3}$	
		c	positive skew	B1(AO1.2) [1]		
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance		
22		a	eg some people have more than one phone oe	E1(AO2.4) [1]	Some of the other figures in the LDS are greater than 1		
		b	E.g. data missing E.g. removal of outlier data items	B1(AO1.1b) B1(AO1.1b) [2]			
		c	Upper quartile	B1(AO1.2) [1]			
		d	$X \sim N(996, 407^2)$, $p = 0.75$ 1271	M1(AO3.4) A1(AO1.1b) [2]	BC allow awrt 1300	if M0 allow B2 for awrt 1300 unsupported	
		e	p -value = 0.127 > 0.05 not significant Insufficient evidence to suggest that the population mean of the number of mobile phones/1000 population has increased	M1(AO1.1b) M1(AO1.1b) A1(AO2.2b) E1(AO2.2b) [4]	from screenshot comparison with 0.05 FT		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		f	Sample size of 12 is small compared to the number of countries in the population so inference may be unreliable	B1(AO2.3) [1]	Any sensible comment on the premise that 12 is a relatively small sample size		
			Total	1			

Question			Answer/Indicative content	Marks	Part marks and guidance	
23		a	$0.2 + 0.15 + 0.3 + k + 0.25 = 1$ $k = 0.1$	M1(AO1.1) A1(AO1.1) [2]	<u>Examiner's Comments</u> Candidates generally did this correctly, with many showing exactly the working given in the mark scheme.	
		b	k^2 seen $0.3 \times 0.25 \times 2 + k^2$ 0.16	M1(AO3.1a) M1(AO1.1) A1(AO1.1) [3]	Or $0.3 \times 0.25 \times 2$ seen Allow slip eg omission of 2; or insertion of 2 Ft k from (a) for M1M1 <u>Examiner's Comments</u> This part caused considerable difficulty, with some candidates omitting it, others drawing a possibility space with equally likely outcomes, and others copying information from the table given in the question. Candidates who realised that a total of 6 could be made either from 3 and 3 or else from 2 and 4 sometimes did not realise that 4 and 2 is also possible; others thought the probability of 3 and 3 had to be counted twice.	
			Total	5		

Question		Answer/Indicative content	Marks	Part marks and guidance		
24		$H_0: p = 0.62$ $H_1: p > 0.62$ p is the proportion of adults over 65 in the (UK population) who use the online social media platform $1 - P(X \leq 45) = 0.0068(1)$ $0.0068 < 0.01$ Result is significant or "reject H_0 " The evidence suggests that the proportion of adults over 65 (in the UK population) using platform has increased from 62%	B1(AO1.1) B1(AO1.1) B1\$(AO2.5) B1&(AO1.1) M1&(AO1.1) A1(AO2.2b) E1(AO2.4) [7]	Allow null Allow alternative May be seen in hypotheses Allow probability NB from use of $B(59, 0.62)$ Comparison of their 0.0068 with 0.01 or 0.68% with 1%; not allowed from point probability Depends on $B1 \& M1 \&$ Conclusion in context Depends on all other marks except $B1 \&$	Allow for sight of 0.9932, (from 0.9931890 2) Allow 'accept H_1 ' OR Critical region ≥ 46 $B1$ 46 in critical region, $M1$, hence conclusion	
				Examiner's Comments The majority of candidates made valid attempts at this question. Common errors at the start of the question included defining H_1 as $p <$		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					0.62 and failing to define p. Some candidates wrongly used the probability that $X \leq 46$ in the test, others wrongly used a point probability. Most candidates who arrived at the correct p value correctly compared it with 0.01 and went on to reject H_0. A small number of otherwise successful candidates did not give a conclusion in the context of the question. There were a small number of scripts from candidates who either omitted the question completely, or wrote vague comments such as 46 out of 59 is about 78%, which is a lot more than 62%.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
25		a	use of $B \sim (20, 0.7)$ soi 0.191638982753...rounded to 2 or more dp isw BC	M1 (AO 3.1b) A1 (AO 1.1) [2]	NB 0.1916 or 0.192 or 0.19 <u>Examiner's Comments</u> This was answered well when clear working was shown. Mistakes may have been due to going straight to the calculator and mistakenly finding at least 14 rather than exactly 14.	
		b	$P(X \leq 13)$ found soi 0.608009811813...rounded to 2 or more dp isw BC	M1 (AO 3.1b) A1 (AO 1.1) [2]	NB 0.3919 90188187 NB 0.6080 or 0.608 or 0.61 M0 if $P(X = 13)$ used NB 0.1643... if M0 allow SC1 for $1 - P(X \leq 14)$ $= 1 -$ 0.58362... $=$ 0.4163708 3 rounded to 2 or more dp <u>Examiner's Comments</u> Candidates who did well in this question made efficient use of the binomial distribution function in their calculator. Candidates who did less well mistook "at least" for "greater than" in this part or made arithmetic slips when using traditional methods to calculate the probabilities.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
26		a	$[\mu =]19$	B1 (AO 1.1) [1]	Examiner's Comments Candidates who did well in this question wrote down the mean by symmetry.	
		b	$1.96 = \frac{21.548 - 19}{\sigma}$ $[\sigma =]$ awrt 1.3 $[\sigma^2 =]$ awrt 1.69	M1 (AO 3.1a) A1 (AO 1.1) A1 (AO 1.1) [3]	$o^{-1.96} = \frac{16.452 - 19}{\sigma}$ r may be implied by final answer allow B3 for awrt 1.69 unsupported NB 1.9599 63985...rou nded to 3 or more sf M0 if z = 2 Examiner's Comments Candidates who did well on this part made efficient use of the standard Normal variable to find the variance. Candidates who did less well confused the mean with the variance, or used a wrong value for z (usually 1.645) to find the variance.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	(i) $[\mu =] 4 \times \text{their } 19 + 5$ $[\sigma^2 =] 4^2 \times \text{their } 1.69$ or $\sigma = 4 \times \text{their } 1.3$ $[Y \sim] N(81, 5.2^2)$ oe	M1 (AO 2.1) M1 (AO 1.1) A1 (AO 1.1) [3]	NB 27.04		
			(ii) 0.04175 or 0.0417 or 0.042 BC	B1 (AO 1.1) [1]	NB 0.0417 462427103		
			Total	8			

Question			Answer/Indicative content	Marks	Part marks and guidance		
27		a	$k(1 \times 6 + 2 \times 5 + 3 \times 4 + 4 \times 3 + 5 \times 2 + 6 \times 1) = 1$ oe $[k =] \frac{1}{56}$ isw	M1 (AO 3.1a) A1 (AO 1.1) [2]	allow one slip in arithmetic B2 if unsupported		
		b	$(6 \times k)^3 \times (12 \times k)^2$ oe seen $\frac{243}{4302592}$ or 0.000056477584 rounded to 2 or more sf	M1 (AO 2.1) A1 (AO 1.1) [2]	FT their k		
		c	$40 \times 6k$ 4.286 or 4.29 or 4.3	M1 (AO 3.1b) A1 (AO 3.2b) [2]	FT their k mark the final answer		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance		
28		a	calculation of $P(X < 14)$ and $P(X > 18)$ 0.3085 and 0.0668 to 1 sf or better these figures do not support the model	M1 (AO 3.4) A1 (AO 1.1) A1 (AO 3.5a) [3]	or solves $-1.476 = \frac{14 - \mu}{\sigma}$ an $0.496 = \frac{18 - \mu}{\sigma}$ d simultaneously $\mu \approx$ and σ $17 \approx 12.02$ 17 is (relatively) far from 15 so not a good fit the second A1 is only available if the first A1 is awarded allow SC2 for showing the model is not a good fit for either value with all working correct or for a complete argument based on symmetry which refers to both tails	or solves $-1.476 = \frac{x - 15}{2}$ an $0.496 = \frac{x - 15}{2}$ d $x = 12.048$ and 15.992 to nearest whole number or better which are not close to 14 and 18 $\frac{14 - 15}{2}$ and $\frac{18 - 15}{2}$ evaluated – 0.5 and 1.5 obtained which are not close to –1.476 and 0.496 respectively	
					<u>Examiner's Comments</u>		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					Candidates who did well in this question made appropriate calculations, showed full working and explained the significance of their results. Candidates who did less well gave incomplete reasoning and/or made slips in their calculations.	
		b	$\Phi^{-1}(0.07) = -1.476 = \frac{14-\mu}{2}$ $[\mu = 16.95]$ OR $\Phi^{-1}(0.69) = 0.496 = \frac{18-\mu}{2}$ $[\mu = 17.008]$ $[\mu =]17$	M1 (AO 3.5c) A1 (AO 2.4) [2]	<i>alternative y</i> since the variance is assumed to be correct, the mean must be as far above the midpoint as it was previously below it. 16 + 1 = 17 if M0 allow B2 for 17 unsupported	

Examiner's Comments

Candidates who did less well in this question ignored the request for an answer correct to two significant figures.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	$z = \pm 1.96$ used $\frac{16 - \mu}{\frac{\sigma}{\sqrt{n}}} < -1.96$ or $\frac{\mu - 16}{\frac{\sigma}{\sqrt{n}}} > 1.96$ $\sqrt{n}$ isolated $\frac{16 - \mu}{\frac{\sigma}{\sqrt{n}}} < -1.96$ oe from their $[n >] 15.3664 - 15.4$ $n = 16$ cao	B1 (AO 1.1a) M1 (AO 3.1b) M1 (AO 2.1) A1 (AO 3.4) A1 (AO 2.2b) [5]	allow method marks only if other z – value, eg – 1.645 used; FT μ eg $\sqrt{n} > 2 \times 1.96$ previous A1 must be awarded for the award of final A1	NB 1.9599 63985... rounded to 3 or more sf M0 if other value for σ used all marks are available if works with = instead of < or > throughout, but withhold final A1 if works with < instead of > or > instead of < throughout
					Examiner's Comments Candidates who did well in this question recognised that the hypothesis test was two-tailed and worked with $\sqrt{n}$ in their calculations. Candidates who did less well worked with an incorrect z value or an incorrect form of the standard deviation. Exemplar 2 $P(X < 14) = 0.07$ $P(X > 15) = 0.11$ $\frac{14 - \mu}{\frac{\sigma}{\sqrt{n}}} = -1.67579$ $\frac{15 - \mu}{\frac{\sigma}{\sqrt{n}}} = 1.3$ $\frac{14 - 15}{\frac{\sigma}{\sqrt{n}}} = -0.5$ $\frac{15 - 15}{\frac{\sigma}{\sqrt{n}}} = 0$ So our model is not a good fit as the z values are far from the actual probabilities. $\frac{14 - \mu}{\frac{\sigma}{\sqrt{n}}} = -1.67579$ $-\mu = -2.95176 - 14$ $\mu = 16.95 = 17$ $\mu = 17$	

Question			Answer/Indicative content	Marks	Part marks and guidance
					$\mu \sim N(17, 2^2)$ $H_0: \mu = 17$ $H_1: \mu \neq 17$ where μ is mean journey time. $n = 16 < 17$ as you want smallest value, find lower tail. $\alpha = 0.05$ $16 - 17 = -1.6667$ z value of $0.025 = -1.96$ $\frac{-1.6667}{\frac{2}{\sqrt{16}}} = -1.6667$ $\frac{16 - 17}{\frac{2}{\sqrt{16}}} \leq \frac{z}{\frac{2}{\sqrt{16}}}$ $n \leq \left(\frac{z}{0.507956} \right)^2$ $-1.6667 \leq \frac{z}{0.507956}$ $n \leq 10.82$ $n = 10$ as n is 100 $n = 10$ large $n = 10$ Note that in part (a) the candidate has chosen a less frequently seen approach, identified in the RH column of the mark scheme. Part (b) is well done. In part (c) the wrong choice of z proved costly.
			Total	10	

Question			Answer/Indicative content	Marks	Part marks and guidance		
29		a	$y = \frac{e^x}{1 - e^x}$ $y(1 - e^x) = e^x$ $[y = e^x(1 + y)] \quad e^x = \frac{y}{1 + y}$ $f^{-1}(x) = \ln\left(\frac{x}{1 + x}\right)$	M1 (AO 1.1a) A1(AO 1.1) A1(AO 2.1) [3]	Clearing fractions Expression for e^x Condone 'y = ' Condone no brackets or mod	x and y may be interchanged at any stage	
		b	$f^{-1}(x) \neq 0$	B1 (AO 1.2) [1]	Allow $y \neq 0$ but not $x \neq 0$		
			Total	4			

Mark Scheme
EXAM PAPERS PRACTICE

Question			Answer/Indicative content	Marks	Part marks and guidance	
30		a	$k(1 \times 2 + 2 \times 3 + 3 \times 4 + 4 \times 5 + 5 \times 6) = 1$ $k = \frac{1}{70}$	M1 (AO1.1) A1 (AO1.1) [2]		
		b	$\frac{2+6+12}{70}$ $= \frac{2}{7}$	M1 (AO1.1) A1 (AO1.1) [2]	FT their k	Allow 1. $P(X \leq 3)$ method
			Total	4		
31			$H_0: \mu = 56$ $H_1: \mu > 56$ μ is the population mean time taken by Statto to complete his walk Use of $N(56, \frac{4^2}{19})$ 0.9459 BC $1 - 0.9459 > 0.05$ No evidence to reject H_0 at 5% level oe There is no evidence to suggest at the 5% level that the mean time taken by Statto to complete his walk has increased	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO1.1) M1 (AO3.4) A1 (AO1.1) E1 (AO2.2b) [7]	Both hypotheses FT their probability	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance		
32		a	0.6	B1 (AO1.1) [1]	BC		
		b	each egg has the same (constant) probability of being cracked	E1 (AO2.4) [1]	OR the probability of any particular egg being cracked is independent of any other egg being cracked or		
		c	$np = \text{their } 0.6$ $p = 0.1$	M1 (AO3.1b) A1 (AO1.1) [2]			
		d	159.4, 106.3, 29.5, 4.4, 0.4, 0.0, 0.0	M1 (AO3.4) M1 (AO1.1) A1 (AO1.1) [3]	Use of B(6, 0.1) so Multiplication by 300 All correct	If unsupported, allow B3 for all frequencies correct, B2 if one error, B1 if two errors	
		e	The theoretical frequencies are close to the observed values, so Rose's model is a good fit.	B1 (AO3.2b) [1]			

	15	
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Question			Answer/Indicative content	Marks	Part marks and guidance	
33		a	eg median $\approx$ mean eg extreme values and/or quartiles symmetrical about mean oe	B1 (AO 2.4) [1]		
		b	$P(X < 46.4) = 0.24956... \approx 0.25$ $P(X < 66.1) = 0.75262... \approx 0.75$ So both values are consistent	M1 (AO 3.3) A1 (AO 2.4) [2]	BC Must follow from correct calc. values	or invNorm(0.25, 56.2, 14.5) = 46.4199... $\approx$ 46.4 i nv Norm(0.75, 56.2, 14.5) = 65.98... $\approx$ 66.1
		c	0.00987574855 ...	B1 (AO 1.1) [2]	BC to 3 or more sf	
		d	Use of $N(112.4, 29^2)$ soi 0.780065376569...	M1 (AO 3.1a) A1(AO 1.1) [2]	BC to 3 or more sf	
		e	thei $\frac{0.78006...}{0.00987... + 0.78006...}$ r 0.98750...so approximately a 99% chance the phone is Wilsons so their father's statement is justifiable	M1 (AO 2.1) A1(AO 2.2b) [2]	their $\frac{0.00987...}{0.00987... + 0.78006...}$ 0.0125...so approximately a 1% chance the phone is Chidera's so their father's statement is justifiable	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance																													
34		a	$k(25 + 9 + 1 + 1 + 9 + 25) = 1$ $k = \frac{1}{70}$	M1 (AO 3.3) A1(AO 1.1) [2]																														
		b	Roxannes model <table border="1" style="margin: 10px auto;"> <tr> <td>day</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr> <tr> <td>f_e</td><td>7</td><td>7</td><td>7</td><td>7</td><td>7</td><td>7</td></tr> </table> Correct for 1 and 4 days, but otherwise a poor fit Alexs model <table border="1" style="margin: 10px auto;"> <tr> <td>day</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr> <tr> <td>f_e</td><td>15</td><td>5.4</td><td>0.6</td><td>0.6</td><td>5.4</td><td>15</td></tr> </table> The expected frequencies are all within 2 of the observed frequencies, so Alex's model appears to be a good fit.	day	0	1	2	3	4	5	f_e	7	7	7	7	7	7	day	0	1	2	3	4	5	f_e	15	5.4	0.6	0.6	5.4	15	B1 (AO 3.4) B1(AO 3.5a) M1(AO 3.4) A1(AO 1.1) B1(AO 2.4) B1(AO 3.5a) [6]	A0 if expected frequencies are rounded to nearest integer	
day	0	1	2	3	4	5																												
f_e	7	7	7	7	7	7																												
day	0	1	2	3	4	5																												
f_e	15	5.4	0.6	0.6	5.4	15																												
			Total	8																														

Question		Answer/Indicative content	Marks	Part marks and guidance		
35		$\bar{X} = 130$ seen $H_0 : \mu = 142$ $H_1 : \mu < 142$ μ is the population mean length of time of meetings use $N\left(142, \frac{17.4^2}{9}\right)$ of $P(\bar{X} < 130) = 0.01927...$ $0.01927... < 0.05$ So [reject H_0], there is evidence at the 5% level to suggest that the mean length of meeting time has decreased.	B1 (AO 1.1) B1(AO 1.1) B1(AO 2.5) M1(AO 3.3) A1(AO 1.1) M1(AO 3.4) A1(AO 2.2b) [1]	BC soi BC FT comparison of their probability		
		Total	7			

Question			Answer/Indicative content	Marks	Part marks and guidance		
36		a	$\frac{1}{24} + \frac{23}{24} \times \frac{1}{23} + \frac{23}{24} \times \frac{22}{23} \times \frac{1}{22}$ $= \frac{1}{8}$	M1 (AO 2.1) M1(AO 2.4) M1(AO 1.1) [3]	$\frac{1}{24}$ seen addition of three terms		
		b	Use $B(24, \frac{1}{8})$ of 0.139(009338...) to 3, 4 or 5 sf	M1 (AO 3.3) M1(AO 1.1) [2]			
		c	$\left(\frac{1}{8}\right)\left(\frac{7}{8}\right)^{23}$ $(0.0057955\dots)^2 = 0.000033588489\dots$ to 3, 4 or 5 sf	M1 (AO 3.1a) A1(AO 1.1) [2]	soi NB 0.00 57955....		
		d	$[6000 \div 120] = 50$ $[1 - 0.13900\dots]^{49} \times 0.0057955\dots$ $0.00000376653\dots$ so Mark is wrong – it's more like 4 in a million	B1 (AO 3.1b) M1(AO 3.1a) A1(AO 1.1) A1(AO 3.2a) [4]	FT their part (b)		
			Total	11			

Question		Answer/Indicative content	Marks	Part marks and guidance	
37	a	$0.2 + 0.1 + k + 2k + 4k = 1$] soi $[k =] 0.1$	M1 (AO1.1a) A1 (AO1.1) [2]	<u>Examiner's Comments</u> Candidates who did well in this question had a clear understanding of random variables and were able to complete the arithmetic successfully. Candidates who did less well made arithmetic slips in one or both parts. A small minority of candidates multiplied the probabilities by the x-values in this part and made no progress in below part.	
	b	$1 - 2 \times \text{their } k$ $0.8 \text{ or } \frac{4}{5} \text{ cao}$	M1 (AO1.1) A1 (AO1.1) [2]	$1 - \text{their } P(X = 4) \text{ oe}$ <u>Examiner's Comments</u> Arithmetic slips were also seen in this part. Those candidates who multiplied the probabilities by the x-values in the part above were unable to progress with the second part of the question.	
		Total	4		

Question		Answer/Indicative content	Marks	Part marks and guidance		
38	a	0.4×0.4 or 0.6×0.2 seen $0.4 \times 0.4 + 0.6 \times 0.2$ 0.28	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) [3]	or tree diagram with correct outcomes and probabilities shown	allow $0.4 \times 0.4 \times 0.4$ or $0.4 \times 0.6 \times 0.2$ 2 mark the final answer	
<u>Examiner's Comments</u> Candidates who did well in this question realised the need to work with P(LW) and P(WW). Many successfully went on to earn all 3 marks, although a few made slips with the arithmetic. Candidates who did not do well incorporated the probability of winning in week 4 – often this was arbitrarily assigned the value of 0.5 and some thought they needed to work with the probabilities for the first three weeks as well.						

Question			Answer/Indicative content	Marks	Part marks and guidance		
	b		$\frac{1}{120}$ or $\frac{5}{120}$ seen $\frac{1}{24}$ or 0.041666...to 2 or more sf	M1 (AO3.1a) A1 (AO1.1) [2]	B2 for 0.041666... unsupported B0 for 0.042 unsupported	mark the final answer	
Examiner's Comments Candidates who did well in this question understood that the desired probability is simply $5 \times \frac{1}{120}$, although a few obtained the correct answer from working out $1 - P(\text{Alf doesn't win})$. Candidates who did less well often found $\frac{1}{24}$ successfully, but then went on to use it in the binomial distribution.							

Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		$1 - \text{their } \frac{1}{24} \text{ evaluated}$ $1 - \left(\frac{23}{24}\right)^n > 0.95 \text{ FT}$ $\left(\frac{23}{24}\right)^n < 0.05 \text{ FT}$ $n = 71 \text{ cao}$	M1 (AO3.1a) M1 (AO2.1) M1 (AO1.1) A1 (AO2.2a) [4]	allow use of =, ≥ or ≤ allow use of = or ≤ 70.3890... unsupported but rounded to 1 or more dp implies M1M1M1	award full marks for 71 unsupported or from trial and improvement	
					Examiner's Comments Candidates who did well in this question subtracted their answer to the above part from 1 and then compared this value to the power of n with 0.05. Often the marks given were FT marks for working with an incorrect probability. Many used trial and improvement instead of logarithms to achieve their final answer. However, the majority of candidates were unable to make meaningful progress with this question, often working with $n \times$ their p .		
			Total	9			

Question			Answer/Indicative content	Marks	Part marks and guidance	
39	a		[=]153	B1 (AO1.1) [1]	<u>Examiner's Comments</u> Most candidates did well in this question and simply wrote down the correct answer. Candidates who did not do well usually wrote the answer 168.	
	b		$z = \pm 1.645$ used their positive $1.645 = \frac{183 - \text{their } 153}{\sigma}$ oe (= 18.237...to 18.248...) $\sigma = 18.2$ cao	B1 (AO1.1) M1 (AO2.1) A1 (AO1.1) [3]	or $\pm 1.644(85..)$) M0 if continuity correction used	<u>Examiner's Comments</u> Candidates who did well in this question used the standard Normal distribution in conjunction with the given information to obtain a value for the standard deviation, giving their answer to the specified precision. Candidates who did less well worked with either an incorrect z-score or an incorrect mean. Sometimes they gave their answer to a different precision.

Question		Answer/Indicative content	Marks	Part marks and guidance		
	c	$\left[\frac{16}{452} = \right] 0.035(398 \dots)$ their $P(X < 120)$ from $N(\text{their } 153, \text{their } 18.2^2)$ probability of 0.0349 to 0.0352 which agrees to 2 sf	B1 (AO3.1b) M1 (AO1.1) A1 (AO2.2b) [3]	allow percentage M0 if continuity correction used allow percentage ----- --- or B1 for their $P(X < 120)$ then M1 for 452 $\times$ their 0.03490.. then A1 for 15.77 to 15.91 which is close to 16 oe ----- --- or B1 as main scheme then M1 for $\sigma = \frac{120 - 153}{\text{their } -1.809}$ and 18.3 $\approx$ 18.2 for A1	or B1 as main scheme then Invnorm(0.0353, their 153, their 18.2) for M1 NB 119.95 to 120.15 which is close to 120 oe for A1 or B1 as main scheme then $z = \frac{120 - \text{their } 153}{\text{their } 18.2}$ for M1 NB -1.809 to -1.813 and Invnorm(0.0353, 0, 1) to obtain -1.806 to -1.812 which is close to -1.809 to -1.813) for A1	
				Examiner's Comments Candidates who did well in this question used their model to obtain a value for $P(X < 120)$ and then made a sensible comparison with $16/452 = 0.035\dots$ They		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					made efficient use of their calculators.	
					Candidates who did less well made errors in calculating their probabilities, sometimes transposing μ and σ . Sometimes they did not communicate their reasoning clearly.	
	d		$H_0: \mu = \text{their } 153$ $H_1: \mu < \text{their } 153$ μ is the population mean flight time from Magaluf to Liverpool use of $N(\text{their } 153,$ $\frac{\text{their } 18.2^2}{24})$ to find $P(\bar{X} < 143.6)$ awrt 0.0057 to 0.0058.. to 2 or more sf oe isw their 0.0057 correctly compared with 0.01 oe result is significant or reject H_0 or accept H_1 there is sufficient evidence to suggest at the 1% level that the mean flight time from Magaluf to Liverpool is less than 153 / mean flight	B1 (AO1.1) B1 (AO2.5) M1* (AO3.3) A1 (AO1.1) M1dep* (AO3.4) A1 (AO1.1) A1 (AO2.2b) [7]	for both hypotheses ; may be stated in words, but need to see 153 for B1 and population mean for 2 nd B1 or inv Norm(0.01, their 153, $\frac{\text{their } 18.2^2}{24})$ $\bar{X} < 144$ to 144.4 is critical region 143.6 correctly compared with their 144.36 FT their comparison FT their comparison	B0 if other parameter used unless clearly defined as population mean or $z = \frac{143.6 - \text{their } 153}{\frac{\text{their } 18.2}{\sqrt{24}}}$ $z = -2.5248..$ to $-2.5302..$ to 2 or more dp their z compared with -2.326 or -2.33 do not allow eg conclude / prove / indicate or other

Question			Answer/Indicative content	Marks	Part marks and guidance	
			time from Liverpool to Magaluf		assertive statement instead of suggest	
					Examiner's Comments Candidates who did well in this question set out their hypotheses clearly in terms of μ, going on to define it as the population mean flight time from Magaluf to Liverpool. They clearly showed how they achieved their probability, making a clear comparison with the significance level. They went on to give a clear conclusion in context. Candidates who did less well gave incomplete / incorrect definitions and conclusions. Candidates who did badly used an incorrect parameter without defining it clearly and / or used continuity corrections and / or worked with $n = 452$ or $n = 1$. Exemplar 3 $\bar{X} = 143.6$ (if μ be the mean flight time from Magaluf to Liverpool) $n = 94$ $\sigma = 18.2$ $X \sim N(153, 18.2^2)$ $\bar{X} \sim N(153, \frac{18.2^2}{94})$ $H_0: \mu = 153$ $H_1: \mu < 153$ 1% s.c. $P(X < 143.6) = 0.005699$ 4 s.p. $\therefore 0.005699 < 0.01$, there is sufficient evidence to reject H_0, i.e. the mean flight time from Magaluf to Liverpool is less than the mean flight time from Liverpool to Magaluf. The hypotheses are correctly set up but the definition of μ is incomplete,	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					so the second B1 is not earned. The rest of the work is good, apart from the final conclusion, as there is no element of suggestion that the mean flight time from Magaluf to Liverpool may be lower than the mean flight time from Liverpool to Magaluf – it is too assertive.	
	e		reduce the value of μ increase the value of σ or σ^2	B1 (AO3.5c) B1 (AO3.5c) [2]	allow B1 for eg use new sample data to calculate new estimate for σ or σ^2 eg take a bigger sample is insufficient Examiner's Comments Candidates who did well on this question considered how the mean and the standard deviation might be adjusted to improve the model. Candidates who did badly wrote in generalities – “collect more data” was commonly seen.	
			Total	16		

Question			Answer/Indicative content	Marks	Part marks and guidance		
40			$\mu = 35$ soi $\sigma = 4$ seen use of $N(35, 4^2)$ to obtain a value for $P(30 < X < 40)$ $[cdfNormal(30, 40, 35, 4) =]$ 0.788700... BC to 2 or more dp use of $Y \sim B(50, \text{their } 0.7887)$ to find $P(Y \leq K)$ or $P(Y \geq K)$ $[1 - P(Y \leq 44)] = 0.032$ to 0.034 BC	B1 (AO1.2) B1 (AO3.1b) M1 (AO2.1) A1 (AO1.1) M1 (AO3.1b) A1 (AO1.1) [6]	by symmetry $35 - 31 = 4$ eg cdfBino mial(50,0.7 887,45,50)	may be embedded in $N(35, 4^2)$ or $N(35, 16)$ M0 if continuity corrections used their 0.7887 must be from use of $N(35, 4^2)$	
					Examiner's Comments Candidates who did well in this question knew the connection between the point of inflection and the standard deviation. They worked with the correct Normal distribution to find the correct probability before using the associated binomial distribution, usually successfully. Candidates who did less well found μ and σ correctly but then either used continuity corrections or divided σ by 50 before using the Normal distribution. A few candidates found 0.7887 correctly and then were unable to make progress. Candidates who did not do well in this question were often unable to progress beyond $\mu = 35$.		
			Total	6			

Question		Answer/Indicative content	Marks	Part marks and guidance		
41	a	$\left[\frac{19.8}{36} =\right] 0.55$	B1 (AO1.1) [1]	or $\frac{11}{20}$ Examiner's Comments Most candidates did this correctly, with some candidates using the values 19 800 000 and 16 200 000 rather than the easier values of 19.8 and 16.2.	Must be dec or simplified frac; but condone 55%	
	b	0.0380(96943116...) BC	B1 (AO1.1) [1]	Correct to at least 2 sf Examiner's Comments Most candidates used their calculators correctly to find the answer. A common wrong answer was 0.28 (from $7 \div 25$).	Allow truncated	
	c	$H_0: p = 0.45$ $H_1: p < 0.45$ p is the probability that a driver speeding / caught speeding / on speed awareness course is female Use or sight of B(25, 0.45) $P(X = 7) = 0.06385(0773831)$ $0.06385 > 0.05$ Result is NOT significant or "accept H_0 " There is insufficient evidence (at the 5% level)	B1 (AO1.1) B1 \$(AO2.5) M1 *(AO3.3) A1 (AO1.1) *M1 (AO3.4) B1 (AO1.1) E1 (AO2.2b)	$O H_0: p = R 0.55$ $H_1: p > 0.55$ p is the probability that a driver speeding / caught speeding / on speed awareness course is male OR Use or sight of B(25, 0.55) $P(X = 18) = 1 - P(X =$	Both hypotheses Must not be inconsistent with H_1 Can be implied by $P(X = 7)$ (OR $P(X = 18) = 0.038..$	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			to suggest that drivers speeding / caught speeding / on speed awareness course are more likely to be male. oe	[7]	$17) = 1 - 0.936149 = 0.06385$ FT their probability Dep on 0.06385 Dep all previous marks except B1\$ SC: Use of $p = 0.5$ can score B1B1M1 (can be implied by 0.0216 or 0.0143) A0M1B0E0 May be incorporated in conclusion Not assertive		
					<u>Examiner's Comments</u> Good answers were either based on females, using $B(25, 0.45)$ and the hypotheses $H_0: p = 0.45$ and $H_1: p < 0.45$, or males, using $B(25, 0.55)$ and the hypotheses $H_0: p = 0.55$ and $H_1: p > 0.55$. Most candidates using a correct value of p then went on to compare 0.0639 ($P(X = 7)$ or $P(X = 18)$) with 0.05. Most then correctly stated that the result is not significant, or that H_0 should not be rejected, and then went on to give an appropriate conclusion in context. However, even good candidates often defined p incorrectly as the probability that a female (or male) driver was speeding, rather than the correct		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					definition as the probability that a driver speeding (or caught speeding) was female (or male). Partial credit was allowed for candidates who used the value of 0.5 for p. There were a considerable number of candidates who made little, or no, progress on this question. Exemplar 3 $Let p = \text{proportion of drivers at speed awareness course who are female}$ $H_0: p = 0.45$ $H_1: p < 0.45$ $X \sim B(25, 0.45)$ $P(X \leq 9) = 0.06385$ $As 0.06385 > 0.05$, we accept the H_0. There is not evidence to suggest male drivers are more likely to exceed the speed limit than female drivers. This candidate is set up and concluded the hypothesis test nicely. The conclusion is non-assertive and in context.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
	d		eg Each driver speeding / caught speeding has the same (constant) probability of being selected for the speed awareness course.	E1 (AO2.4) [1]	OR the probability of any particular driver caught speeding being selected for the speed awareness course is independent of any other driver caught speeding being selected.		
					Examiner's Comments This part was often omitted or generic. The context of the question needed to be considered in the answer.		
					Exemplar 4 <u>Events occurred mutually and independently.</u> The candidate has offered up the idea of independence, but has not used this in the context of the question looking at the probability of any one driver caught speeding being selected is independent of any other driver caught speeding being selected.		
			Total	10			

Question			Answer/Indicative content	Marks	Part marks and guidance		
42	a		0.0348	B1 (AO1.1) [1]	BC 0.0348247 5	Allow if seen correct to at least 3 sf. We do not want to penalize wrong rounding too often	
					Examiner's Comments Part (a) was done correctly by about half the candidates, with part (b) proving rather more difficult.		
	b		$P(X \leq 4) = 0.013(337876\dots)$ or $P(X \leq 3) = 0.004(09062\dots)$ found Hence 4 is not in CR, or 3 is in CR so CR is $\{0, 1, 2, 3\}$ or $X \leq 3$	M1 (AO3.1a) M1 (AO2.2a) A1 (AO3.2a) [3]	Allow M1 for one (correct) of these. Both probabilities must be correct	Accept $P(X \leq 4) > 0.01$ or $P(X \leq 3) < 0.01$	
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance	
43	a		Quota sampling	B1 (AO1.2) [1]		
	b		9	B1 (AO1.1) [1]	from 5×1.8	
	c		Systematic: select every 24th number on the list start randomly between $n = 1$ and $n \geq 184$ and stop when 200 have been selected (if $n > 184$, must cycle through list) Simple random sampling: assign each item in the list a unique number (eg from 1 to 4960) generate random numbers until a sample of 200 has been selected so	M1 (AO2.4) A1 (AO1.1) E1 (AO2.4) E1 (AO1.1) [4]	<i>alternatively</i> select every 25th number on the list start randomly between $n = 1$ and $n \geq 25$, and cycle through the list again, stopping when 200 have been selected <i>alternatively</i> allow any process where each member of the population has an equal chance of being selected allow any	<i>alternatively</i> select every 24.8th value on list, rounding as appropriate start randomly with any value on list. Cycle through the list repeatedly until 200 items have been selected

Question			Answer/Indicative content	Marks	Part marks and guidance		
					process where each possible sample has an equal chance of being selected		
	d		as the size of the sample increases, the shape of the distribution appears more and more "Normal" oe	B1 (AO2.4) [1]	must refer to shape and closer to Normal shape for larger sample		
	e		use of $N(60.0515, 6.5783^2)$ to find $P(X > 65)$ awrt 0.23 $4960 \times \text{their } 0.226$ 1121 or 1120 or 1119	M1 (AO3.3) A1 (AO3.4) M1 (AO3.1b) A1 (AO3.5a) [4]	condone use of 6.5717^2 or parameters rounded to 3 sf	M0 if continuity correction used or eg $P(X > 64)$ found	
	f		eg there may be seasonal fluctuations such as teachers retiring in August	B1 (AO3.5b) [1]	allow any sensible reason in context	do not allow eg mean and sd may be different	
			Total	12			

Question		Answer/Indicative content	Marks	Part marks and guidance		
44		$H_0: \mu = 5$ oe $H_1: \mu \neq 5$ oe H_1 takes this form as Claud is testing whether the mean length of time is different to 5 oe μ is the population mean time for which computers are kept before being replaced use of $N(5, \frac{2.7054^2}{120})$ to find $P(\bar{X} < 4.8855)$ or $\text{invNorm}(p, 5, \frac{2.7054}{\sqrt{120}})$ where $p = 0.025$ or 0.05; may be implied by 0.3215 or $4.51595\dots$ or $4.59377\dots$ $P(\bar{X} < 4.8855) = \text{awrt } 0.32$ $0.32 > 0.025$ or $4.8855 > 4.5(2)$ not significant or accept H_0 or do not reject H_0 or reject H_1	B1 (AO1.1) B1 (AO2.4) B1 (AO2.5) M1 (AO3.3) A1 (AO1.1) M1 (AO3.4) A1FT (AO1.1) A1FT (AO2.2b) [8]	allow any parameter apart from $\bar{x}$ for population mean as long as clearly defined as (population) mean condone use of 2.6941 (may be rounded) instead of 2.7054 for M1, may be implied by 0.32076 or CR is $\bar{x} < 4.51797$, but A1 not available or CR is $(\bar{X}) < 4.5 - 4.6$ comparison of <i>their</i> probability with 0.025 or comparison of 4.8855 with <i>their</i> critical value from	after B marks M1A1M1A1 A1 may be earned if working with 2.7054 rounded to 2 or more sf $z = \frac{4.8855 - 5}{\frac{2.7054}{\sqrt{120}}}$ For M1 awrt -0.46 A1 (may be implied by -0.466 or $-0.465567\dots$ if 2.6941 used) <i>their</i> – $0.4436 > -1.96$ oe M1 dep on award of previous M1	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			insufficient evidence to suggest (at 5% level) that the (population) mean length of time (computers are kept) is not 5 years		use of 0.025, as long as previous M1 awarded may be embedded in conclusion in context	do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; A0 if answer spoiled	
			Total	8			

Question		Answer/Indicative content	Marks	Part marks and guidance		
45	a	$2p + q + 0.2 + 0.3 = 1$ soi oe $2 \times p \times q = 0.06$ soi eliminate p or q with a correct substitution from one of <i>their</i> equations $q^2 - 0.5q + 0.06 = 0$ or $2p^2 - 0.5p + 0.03 = 0$ oe $q = 0.2$ or 0.3 and $p = 0.15$ or 0.1 $(q < 2p \text{ so}) q = 0.2$ and $p = 0.15$	B1 (AO2.1) M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) A1 (AO3.2a) [6]	allow M1 if 2 omitted eg $2 \times \frac{0.03}{q} + q = 0.5$ or $2p + \frac{0.03}{p} = 0.5$ may be implied by eg $q = 0.2$ or 0.3 and $2p = 0.3$ or 0.2	NB if 2 omitted, A0 for $2p^2 - 0.5p + 0.6 = 0$ or $2q^2 + q + 0.24 = 0$ which have no real root	
	b	$10 \times q \times (1 - q)^9$ soi 0.27 or 0.268 or awrt 0.2684 isw	M1 (AO1.1) A1 (AO1.1) [2]	FT <i>their</i> q where $0 < q < 1$		

Question		Answer/Indicative content	Marks	Part marks and guidance		
	c	$H_0: p = 0.2$ $H_1: p > 0.2$ p is the probability that the spinner shows a 1 (on any given spin) oe use of $X \sim B(100, 0.2)$ where x is the number of 1s obtained in 100 spins to obtain $P(X \geq k)$ or $P(X \leq k)$ $P(X \leq 27) = \text{awrt } 0.97$ or $P(X \geq 28) = \text{awrt } 0.034$ $0.034 < 0.05$ or $0.97 > 0.95$ significant or reject H_0 or accept H_1; may be embedded in conclusion in context there is sufficient evidence to suggest (at 5% level) that the probability of a score of 1 is greater than 0.2	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO1.1) M1 (AO3.4) A1 (AO1.1) A1 (AO2.2b) [7]	both hypotheses ; allow equivalent in words or eg $P(1) = 0.2$ $k = 27, 28$ or 29 or critical region is $X \geq 28$ or 28 is in critical region must have the correct probability or correct critical region for the last two A marks	allow any parameter as long as clearly defined as probability M0 for $P(X = k)$ NB $P(X = 28) = 0.014...$ $P(X = 27) = 0.020168...$ FT <i>their</i> probability, dependent on award of first M1 do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; A0 if answer spoiled	
		Total	15			